

Linear Algebra (MT1004)

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Course Instructor(s)

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Final Exam

Total Time: 3 Hours

Total Marks: 150

Total Questions: 6

3J
Section

Student Signature

Roll No

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Important Instruction:

1. Only simple calculator is allowed. Use of programmable calculator or any helpful devices will result **F** grade in the course.
2. Wrong calculation work found (if any) at a step will not be further marked. Marks will be awarded till the correct calculations.
3. Attempt all question parts together. Question attempted in separate parts will not be marked.
4. **Question # 6, Must be returned to the invigilator along with the answer sheet.**

CLO #1: Use concept of elementary row operations to find the inverse of square matrices, determinant of a matrix and solving the system of linear equations.

CLO #5: Express a linear transformation graphically using matrices and to solve problems.

Question#1: [25]- marks:

- a) [3+4]: Use **Inversion Algorithm** to find the Inverse of a matrix $A = \begin{bmatrix} 1 & -1 \\ 2 & 3 \end{bmatrix}$. Write down A as a product of elementary matrices $A = E_1^{-1}E_2^{-1}E_3^{-1} \dots E_{k-1}^{-1}E_k^{-1}$ for some integer k .
- b) [4]: Verify that $A^{-1} = E_kE_{k-1} \dots E_3E_2E_1$ for some integer k .
- c) [8]: Discuss the **Geometric Effect** on the Unit square by the given matrix A , using the following step:
 - i) Show mathematically and graphically effect of $A = E_1^{-1}E_2^{-1}E_3^{-1} \dots E_{k-1}^{-1}E_k^{-1}$ on the Unit Square with end points at $(0, 0)$, $(1, 0)$, $(1, 1)$, $(0, 1)$ step by step.
- d) [3]: Show that succession of shears, compressions, expansions, and reflections obtained in part (c) produces the same image as obtained by A .
- e) [3]: Find an equation for the image of the line $y = 2x + 1$ under multiplication by the matrix

$$A = \begin{bmatrix} 1 & -1 \\ 2 & 3 \end{bmatrix}.$$

CLO #2: Properties of vectors in 2-space, 3-space and n-space and recognize vector spaces and/or subspaces to compute their bases and its dimension.

Question#2: [15]- marks:

- a) Let $V = \mathbb{R}^2$ be the set of all ordered pairs of real numbers, and consider the following addition and scalar multiplication operations on $u = (u_1, u_2)$ and $v = (v_1, v_2)$:

$$u + v = (u_1 + v_1 - 1, u_2 + v_2 - 1), \quad ku = (ku_1, ku_2).$$

- [2]: Use vector space **Axiom 4** to find a zero vector such that $u + (0) = u$ for $u = (u_1, u_2)$.
 - [2]: Use vector space **Axiom 5** by producing a negative vector $(-u)$ such that $u + (-u) = 0$ for $u = (u_1, u_2)$.
 - [2]: Check whether **Axiom 8**, i.e. $[(k+m)u = ku + mu]$ and **Axiom 10** i.e. $[1 \cdot u = u]$, holds or not?
- b) For the linear systems $AX = B$, and $AX = 0$, determine the solution space X , where

$$B = \begin{bmatrix} 0 \\ -1 \\ 5 \\ 6 \end{bmatrix} \text{ and } A \text{ is given below}$$

$$A = \begin{bmatrix} 1 & 3 & -2 & 0 & 2 & 0 \\ 2 & 6 & -5 & -2 & 4 & -3 \\ 0 & 0 & 5 & 10 & 0 & 15 \\ 2 & 6 & 0 & 8 & 4 & 18 \end{bmatrix}$$

- [2]: Find the **General Solution** of $X = B$, and write the solution space X in the form (Column Matrix Notation) i.e. $X = X_0 + c_1v_1 + c_2v_2 + c_3v_3 + \dots + c_kv_k$. Can you explain what geometrically this equation represents?
- [2]: Find the basis for the Column and Null space of A and explain geometrically these solution spaces.

Hint: Write the solution space say " X " in Matrix column notation and then check cardinality of X , and basis for null space of A .

- [3]: Show that the Null Space of A is a sub-space of $\mathbb{R}^6$.
- [2]: Verify **Dimension Theorem** for A .

CLO #3: Use of Eigen Value/Vectors to diagonalize and orthogonally diagonalize a matrix, and use them in different applications.

Question#3: [35]- marks:

- a) [6]: Define Orthogonal Diagonalization, Algebraic and Geometric Multiplicity of square Matrix A .

- b) [6]: For the Matrix $A = \begin{bmatrix} 0 & 0 & -2 \\ 1 & 2 & 1 \\ 1 & 0 & 3 \end{bmatrix}$, show that the Eigen values are $\lambda_1 = 2$ and $\lambda_2 = 1$, Also find the basis and dimension of the Eigen Spaces corresponding to each Eigen value.

- c) [3]: Use definition of $AX = \lambda X$ to determine eigenvectors of A^7 .
- d) [5]: If A is an $n \times n$ diagonalizable matrix and $P^{-1}AP = D$. Show that $A^k = PD^kP^{-1}$, where P is the matrix consisting of Orthonormal basis for A and k is an integer.
- e) [4]: Find the geometric and algebraic multiplicity of each eigenvalue of the matrix A .
- f) [9]: Determine whether A is diagonalizable or not? If A is diagonalizable, then find a matrix P such that $P^{-1}AP = D$ diagonalizes A .
- g) [2]: Check that matrix A and $P^{-1}AP$ have the same determinant and trace (Hint: Similarity Invariants).

CLO #4: Identify Inner product spaces and/or perform Gram Schmidt process/QR decomposition using inner products.

Question#4: [15]- marks:

Suppose $A = \begin{bmatrix} 1 & -1 & 2 \\ -1 & 1 & 1 \\ 0 & 1 & -2 \end{bmatrix}$, define the column vectors u_1, u_2 and u_3 as

$$u_1 = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}, \quad u_2 = \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix} \quad \& \quad u_3 = \begin{bmatrix} 2 \\ 1 \\ -2 \end{bmatrix}.$$

- a) [7]: Use the Gram – Schmidt process to find an orthogonal set of vectors $\{v_1, v_2, v_3\}$ and then find orthonormal set of vectors $\{q_1, q_2, q_3\}$.
- b) [8]: Using part a) find the matrix Q , such that $A = QR$ where, $Q = [q_1 \mid q_2 \mid q_3]$ and R is given below

$$R = \begin{bmatrix} \langle u_1, q_1 \rangle & \langle u_2, q_1 \rangle & \langle u_3, q_1 \rangle \\ 0 & \langle u_2, q_2 \rangle & \langle u_3, q_2 \rangle \\ 0 & 0 & \langle u_3, q_3 \rangle \end{bmatrix}.$$

Give reason: Why the entries below the main diagonal are always zero in ?

Note: Use standard dot product of vectors as an inner product between vectors.

CLO #5: Express a linear transformation graphically using matrices and to solve problems.

Question#5: [40]- marks:

- a) [5]: Define **Isomorphism** between two vector spaces V and W and give one example. Show that $\begin{bmatrix} a & b \\ c & d \end{bmatrix} \mapsto (a, b, c, d)$ is an isomorphism between $V = M_{2 \times 2}$ and $W = \mathbb{R}^4$.
- b) [3]: Is it true that every n – dimensional vector space is isomorphic to $\mathbb{R}^n$. If your answer is yes, define only isomorphism between $P_n(x)$ and $\mathbb{R}^{n+1}$.
- c) [5]: Define basis for a vector space V . If we think of a basis as describing a coordinate system for a finite dimensional vector space V , then describe the significance of the basis axioms of V .

- d) [2]: Let $T: V \rightarrow W$ be linear transformation with $\dim(V) = n$ and $\dim(W) = m$. Suppose $B = \{u_1, u_2, \dots, u_n\}$ be the basis of V and $B' = \{v_1, v_2, \dots, v_m\}$ be the basis for W . Write a formula for the matrix of T relative to the bases B and B' .

- e) [3+2+5]: Let V be a finite-dimensional vector space, and let B and B' be two bases of V . Suppose $T: V \rightarrow V$ and $I: V \rightarrow V$ be the linear operators then show that:

- i) $P_{B \rightarrow B'} = [I]_{B', B}$ and $P_{B' \rightarrow B} = [I]_{B, B'}$.
 ii) Show that $(P_{B \rightarrow B'})(P_{B' \rightarrow B}) = I$ and conclude that

$$P_{B \rightarrow B'} = (P_{B' \rightarrow B})^{-1}.$$

Hint: Use definition of the Transition Matrix.

- iii) Use the above results to show that the matrix of T relative to the basis B' satisfies $[T]_{B'} = P^{-1}[T]_B P$ where $P = P_{B \rightarrow B'}$ and $P^{-1} = P_{B' \rightarrow B}$.

Hint: Use $T = I \circ T \circ I$, in part d).

- f) [2]: Define Eigen value(s) and Eigen vector(s) of a square matrix.

If $C = \begin{bmatrix} 1 & -1 \\ 0 & 2 \end{bmatrix}$ and $D = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$, then

- a. [4]: Determine Eigen values and Eigen vectors of C .

- b. [9]: Show that C and D represents the same linear operator $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ by showing $P^{-1}CP = D$, where $P = P_{B' \rightarrow B} = \begin{bmatrix} [u'_1] & [u'_2] \end{bmatrix}$ & $P^{-1} = P_{B \rightarrow B'}$, $B' = \{u'_1, u'_2\}$, $B = \{e_1, e_2\}$, $u'_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$, $u'_2 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$. Here P & P^{-1} represents the transition matrices.

$$u'_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad u'_2 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

Bonus Questions: Maximum allowed points are 20.

1. [10]: State ONLY Equivalence Theorem negation statements.

2. [10]: Let $T: R^4 \rightarrow R^3$ be a linear transformation defined as

$$T(a_1, a_2, a_3, a_4) = (a_1 + a_3, a_1 + a_2 + 2a_3, 2a_1 + a_2 + 3a_3).$$

Determine:

- i) Is T onto?
- ii) Find the basis for the range of T i.e. $R(T)$.
- iii) Find the kernel of T i.e. $\ker(T)$.
- iv) Is T one to one?

3. [10]: If $T_1: U \rightarrow V$ and $T_2: V \rightarrow W$ are one to one linear transformation, then show that

- i) $T_2 \circ T_1$ is one to one.
- ii) $(T_2 \circ T_1)^{-1} = T_1^{-1} \circ T_2^{-1}$

4. [10]: Let B and B_1 are the basis for a vector space V and W respectively. Consider $T: V \rightarrow W$ is a linear transformation. Under what condition(s) and how can we prove that $T(B)$ i.e. image of the basis elements of V will form a basis for W ? **Hint:** Assume V and W are finite dimensional vector spaces.

5. [10]: Let $T: V \rightarrow W$ be linear transformation with $\dim(V) = n$ and $\dim(W) = m$. Let $B = \{u_1, u_2, \dots, u_n\}$ be the basis of V and $B' = \{v_1, v_2, \dots, v_m\}$ be the basis for W . Show that $[u_i]_B = e_i$, the i^{th} standard base vector, and that for the matrix A of T relative to B and B' , $A[u_i]_B = [T(u_i)]_{B'}$.

6. [10]: Let $T: V \rightarrow W$ be a linear transformation between finite-dimensional vector spaces. Show that:

- i) If $\dim(W) < \dim(V)$, then T cannot be one-to-one.
- ii) If $\dim(V) < \dim(W)$, then T cannot be onto.

Good Luck